Hotels

Hotel_ID (INT, Primary Key, Auto-increment)

Name (VARCHAR(100), NOT NULL)

Location (VARCHAR(100), NOT NULL)

Rooms

Room_ID (INT, Primary Key, Auto-increment)

Hotel_ID (INT, Foreign Key REFERENCES Hotels(Hotel_ID))

Room_Type (VARCHAR(50), NOT NULL)

price_per_night (DECIMAL(10,2), NOT NULL)

Guests

Guest_ID (INT, Primary Key, Auto-increment)

Name (VARCHAR(100), NOT NULL)

Contact (VARCHAR(20), NOT NULL)

Bookings

Booking_ID (INT, Primary Key, Auto-increment)

Guest_ID (INT, Foreign Key REFERENCES Guests(Guest_ID))

Room_ID (INT, Foreign Key REFERENCES Rooms(Room_ID))

Checkin_Date (DATE, NOT NULL)

Checkout_Date (DATE, NOT NULL)

Create table——--

```sql
-- Hotels
CREATE TABLE Hotels (
  Hotel_ID INT AUTO_INCREMENT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Location VARCHAR(100) NOT NULL
);

-- Rooms
CREATE TABLE Rooms (
  Room_ID INT AUTO_INCREMENT PRIMARY KEY,
  Hotel_ID INT,
  Room_Type VARCHAR(50) NOT NULL,
  price_per_night DECIMAL(10,2) NOT NULL,
  FOREIGN KEY (Hotel_ID) REFERENCES Hotels(Hotel_ID)
);

-- Guests
CREATE TABLE Guests (
  Guest_ID INT AUTO_INCREMENT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Contact VARCHAR(20) NOT NULL
);

-- Bookings
CREATE TABLE Bookings (
  Booking_ID INT AUTO_INCREMENT PRIMARY KEY,
  Guest_ID INT,
  Room_ID INT,
  Checkin_Date DATE NOT NULL,
  Checkout_Date DATE NOT NULL,
  FOREIGN KEY (Guest_ID) REFERENCES Guests(Guest_ID),
  FOREIGN KEY (Room_ID) REFERENCES Rooms(Room_ID)
);
-- Hotels
INSERT INTO Hotels (Name, Location) VALUES
('Ocean View Inn', 'Santa Monica'),
('Mountain Heights Resort', 'Denver'),
('Urban Stay Inn', 'New York'),
('Countryside Cottage', 'Nashville'),
('Lakeside Lodge', 'Lake Tahoe');

-- Rooms
INSERT INTO Rooms (Hotel_ID, Room_Type, price_per_night) VALUES
(1, 'Single', 80.00),
```

```sql
(1, 'Double', 100.00),
(2, 'Single', 90.00),
(2, 'Suite', 150.00),
(3, 'Double', 120.00),
(4, 'Single', 85.00),
(4, 'Double', 105.00),
(5, 'Single', 95.00),
(5, 'Suite', 160.00),
(5, 'Double', 125.00);

-- Guests
INSERT INTO Guests (Name, Contact) VALUES
('Alice Harper', '555-1234'),
('Bob Mitchell', '555-2345'),
('Charlie Kim', '555-3456'),
('Diana Ross', '555-4567'),
('Ethan Scott', '555-5678');

-- Bookings
INSERT INTO Bookings (Guest_ID, Room_ID, Checkin_Date, Checkout_Date) VALUES
(1, 1, '2025-03-01', '2025-03-04'),
(1, 2, '2025-04-10', '2025-04-15'),
(2, 4, '2025-04-05', '2025-04-12'),
(3, 5, '2025-04-01', '2025-04-06'),
(4, 7, '2025-04-20', '2025-04-30'),
(5, 10, '2025-04-18', '2025-04-26'),
(2, 3, '2025-04-07', '2025-04-09'),
(5, 9, '2025-04-04', '2025-04-10'),
(4, 6, '2025-04-06', '2025-04-13'),
(3, 5, '2025-04-15', '2025-04-21');
```

//////////////////////////////////////

Question & Answer:-

1.List all hotels along with their locations.

Solve;-

```sql
SELECT Name, Location
FROM Hotels;
```

2.Display all rooms with a price greater than 100 per night

Solve-

```sql
SELECT Room_ID, Hotel_ID, Room_Type, price_per_night
FROM Rooms
WHERE price_per_night > 100;
```

3. Retrieve the details of guests who made a booking in March 2025
Solve;-
```
SELECT G.Guest_ID, G.Name, G.Contact
FROM Guests G
JOIN Bookings B ON G.Guest_ID = B.Guest_ID
WHERE B.Checkin_Date >= '2025-03-01'
  AND B.Checkin_Date < '2025-04-01';
```

4.Count the total number of rooms in each hotel
Solve;-
```
SELECT Hotel_ID, COUNT(Room_ID) AS total_rooms
FROM Rooms
GROUP BY Hotel_ID;
```

5. Find the hotel with the highest number of booking reservations
Solve;-
```
SELECT TOP 1 H.Hotel_ID, H.Name, COUNT(B.Booking_ID) AS total_bookings
FROM Hotels H
JOIN Rooms R ON H.Hotel_ID = R.Hotel_ID
JOIN Bookings B ON R.Room_ID = B.Room_ID
GROUP BY H.Hotel_ID, H.Name
ORDER BY total_bookings DESC;
```

6.List the names of guests and their respective booking IDs
Solve-
```
SELECT G.Name, B.Booking_ID
FROM Guests G
JOIN Bookings B ON G.Guest_ID = B.Guest_ID;
```

7.Calculate total revenue for each hotel
Solve
```
SELECT H.Hotel_ID, H.Name,
SUM(R.price_per_night * DATEDIFF(B.Checkout_Date, B.Checkin_Date)) AS total_revenue
FROM Hotels H
JOIN Rooms R ON H.Hotel_ID = R.Hotel_ID
JOIN Bookings B ON R.Room_ID = B.Room_ID
GROUP BY H.Hotel_ID, H.Name;
```

8.Display booking details where checkout is after March 10th, 2025
Solve
```
SELECT Booking_ID, Guest_ID, Room_ID, Checkin_Date, Checkout_Date
FROM Bookings
WHERE Checkout_Date > '2025-03-10';
```

9. Retrieve room details for a particular hotel (say, Hotel_ID = 1)
Solve;-

```
SELECT Room_ID, Room_Type, price_per_night
FROM Rooms
WHERE Hotel_ID = 1;
```

10. List the names of guests who stayed more than 5 nights
Solve

```
SELECT G.Name
FROM Guests G
JOIN Bookings B ON G.Guest_ID = B.Guest_ID
GROUP BY G.Guest_ID, G.Name
HAVING SUM(DATEDIFF(B.Checkout_Date, B.Checkin_Date)) > 5;
```